MODEL PROCEDURES FOR CHEMICAL ANALYSIS
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1. Determination of CaO in cement by Complexometric titration

Part A: Preparation of Standard Solution of EDTA

Weigh out the given ethylene diamine tetra acetic acid crystals in to a 100 cm3 volumetric flask. Dissolve in distilled water. Dilute up to the mark and mix well.

$$\text{Molarity of EDTA} = \frac{\text{Weight of EDTA in } 100\text{cm}^3 \times 10}{\text{Gram Molecular Weight of EDTA Salt (372.24)}}$$

Part B: Estimation of CaO

Pipette out 25 cm^3 of the given sample solution a clean conical flask using a dispenser. Add 5 cm^3 of diethylamine, followed by 10 cm^3 of 4N Sodium hydroxide solution. Add about 10 drops of polyvinyl alcohol and a pinch of Patton & Reeder's indicator. Titrate the solution against EDTA solution taken in the burette until the color changes from wine red to pure blue. Note down the burette reading and repeat the titration to get concordant values.

Burette:	Standard Ethylene diamine tetra acetic acid solution
Conical Flask	25 cm^3 of water sample + 5 cm^3 diethylamine + 10 cm3 of 4N Sodium hydroxide solution
Indicator:	Patton & Reeder's indicator
End Point:	Wine red to pure blue

VIVA QUESTIONS

1. What is cement?
2. How are the constitutions of cement?
3. What is EDTA?
4. Write the structural formula for EDTA.
5. Why is disodium salt of EDTA preferred to EDTA?
6. Why is Ammonia solution added while preparing EDTA solution?
7. 2.0746 g of EDTA was dissolved in 100 cm^3 of water. What is the molar concentration of EDTA?
8. What is buffer solution?
9. What is the pH of the buffer used? How is it prepared?
10. Why is the indicator Patton Redeer's shows wine red color at the beginning and blue color at the end?
11. Name a metal ion sensitive indicator.
12. Why is the titration involving EDTA carried out slowly towards the end point?
13. What is the application of amount of CaO data in environmental engineering practice?
14. How do you explain the color change of solution at the end point of titration?

2. Estimation of percentage of Copper in Brass by Iodometric titration

Part A: Preparation of Brass Solution

Weight out the given brass sample in to 250 cm^3 conical flask. Add concentrated nitric acid drop by drop till it dissolves. Boil till brown fumes are expelled. Add one spatula of urea and one test tube of distilled water. Boil for two minutes. Cool to room temperature.

Part B: Estimation of Copper

Burette:	Standard Sodium thiosulphate solution
Conical Flask:	Brass solution prepared above + Ammonium hydroxide solution till Bluish white precipitate + ¼ test tube of acetic acid to dissolve the precipitate + 1 test tube of 10% Potassium iodide solution.
Indicator:	2 cm^3 of freshly prepared starch solution towards the end
End Point:	Disappearance of dark blue colour.

VIVA QUESTIONS

1. What are the constituents of Brass?
2. How is Brass solution prepared?
3. What is the purpose of adding Urea?
4. What are Iodometric titrations?
5. Why is Ammonium Hydroxide added to the Brass solution?
6. What is the bluish white precipitate formed after adding Ammonia solution?
7. Why is the solution acidified with Acetic acid after the addition of Ammonium Hydroxide?
8. Why is Acetic acid added?
9. How is liberated Iodine estimated?
10. What is the reaction that occurs between iodine and Sodium thiosulphate?
11. Write the reaction between acidified Brass solution and Potassium Iodide.
12. Why is starch indicator added towards the end point?
13. What is the white precipitate produced at the end point?

3. Estimation of Iron in the given sample of Haematite ore solution by Redox titration

Part A: Preparation of standard potassium dichromate

Weigh out the given potassium dichromate crystals in to a 100 cm^3 volumetric flask. Dissolve in distilled water. Dilute up to the mark and mix well.

Normality of $K_2Cr_2O_7$ = $\frac{\text{Weight of } K_2Cr_2O_7 \text{ in 100 } cm^3 \text{ x 10}}{\text{Gram molecular weight of } K_2Cr_2O_7 \text{ (49.03)}}$

Part B: Estimation of iron

Burette:	Potassium dichromate solution
Conical Flask:	25 cm^3 of ore solution + ¼ test tube of 1:1 hydrochloric acid solution. Heat nearly to boiling. Add stannous chloride drop wise till it becomes colorless + 2 drops of $SnCl_2$ extra. Cool to room temperature and add 2 test tube of distilled water. Add 3 cm^3 of mercuric chloride at a stretch. Indicator: Drops of potassium ferricyanide on a paraffin wax paper.
End Point:	No blue or green colouration when a drop of the mixture is placed on the indicator drop.

VIVA QUESTIONS

1. What is the main constitutes of Haematite ore?
2. In what form does Iron exist in Haematite?
3. What is the oxidation number of Iron in Haematite?
4. Give the others form of Iron ore.
5. What is the role of Stannous Chloride?
6. Why is Mercuric Chloride added and at a stretch?
7. What happens when the excess of Stannous Chloride is not removed?
8. What is the indicator used?
9. Why the color of the indicator drop remains the same at the end point?
10. Name any two oxidizing and reducing agents used in the experiment.
11. What is the reaction that occurs during the titration?
12. Write down the equations for the
    a. Reactions between hot acidified ore solution and Stannous Chloride.
    b. Addition of saturated mercuric Chloride to the solution in **a.**
    c. Addition of Potassium Dichromate to the solution obtained in **b.**
13. Why is it necessary to add very little excess of Stannous Chloride during the reduction of Fe^{3+} to Fe^{2+}?
14. Why grey ppt. is obtained some time?
15. Why Stannous Chloride is added while the solution is hot?

4. Determination of Chemical Oxygen Demand (COD) of the given industrial waste water sample

Part A: Preparation of standard ferrous ammonium sulphate (FAS)

Weigh out the given ferrous ammonium sulphate crystals in to a 100 cm^3 volumetric flask. Add one test tube of dilute sulphuric acid solution. Dissolve in distilled water. Dilute up to the mark and mix well.

$$\text{Normality of FAS} = \frac{\text{Weight of EDTA in 100 } cm^3 \times 10}{\text{Gram molecular weight of FAS (392.14)}}$$

Part B: Estimation of COD

Burette:	Ferrous ammonium sulphate solution
Conical Flask:	25 cm^3 of wastewater + 25 cm^3 of Potassium dichromate solution + 1 test tube of dil H_2SO_4
Indicator:	3 drops of ferroin
End Point:	Bluish green to reddish brown

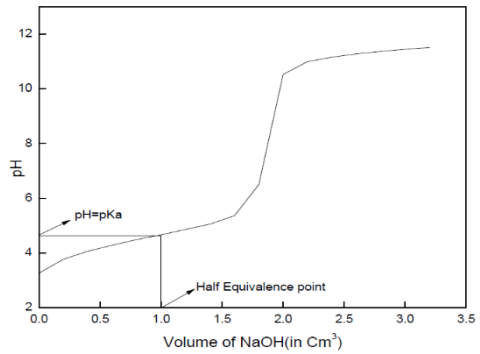
Part C: Blank titration is done as above method without waste water sample.

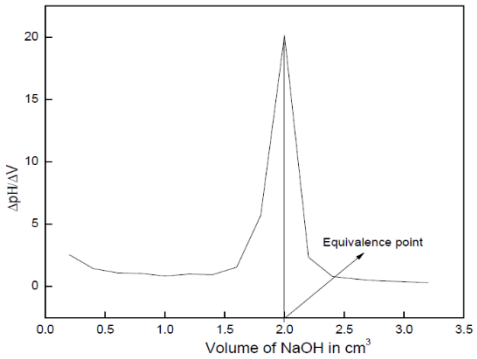
VIVA QUESTIONS

1. What is chemical Oxygen demand?
2. What are the constituents which increase the COD of sewage?
3. What general group of organic compounds is not oxidized in the COD test?
4. What is the role of Silver Sulphate?
5. What is the role of Mercuric Sulphate?
6. Explain the role of acidified Potassium Dichromate.
7. What are the products formed after COD analysis?
8. Why is Sulphuric acid added during the preparation of standard FAS solution?
9. Why can't we use Nitric acid instead of dilute Sulfuric acid?
10. What is the composition of Ferroin?
11. What is the unit of COD?
12. Mention a few applications of COD test in environmental engineering practice.
13. What is the limitation of COD?
14. What conclusion can be drawn from the COD measurement?

5. Determination of pKa value of weak acid

Pipette out 50cm^3 of the given weak acid into a beaker. Immerse the glass electrode-calomel electrode assembly into it. Connect the electrodes to pH meter and measure the pH. Add sodium hydroxide from the burette in the increments of O.2 cm^3 and measure the pH after each addition. Plot a graph of ΔpH/ ΔV against volume of sodium hydroxide solution added and determine the equivalence point. Plot another graph of pH against volume of sodium hydroxide solution added and determine the pKa value of the weak acid.



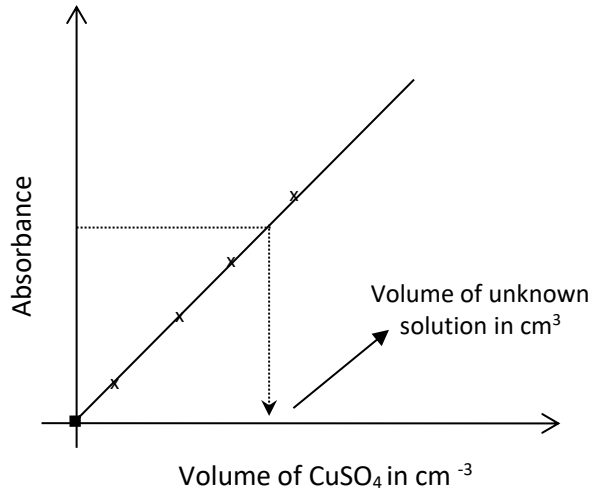


VIVA QUESTIONS

1 Define pKa of a weak acid.
2 pK_a values of two acids are 2.8 and 3.2. Which one of these acids is stronger?
3 Why should we compute the pKa values of a weak acid?
4 Define pKa of a weak acid in terms of pH of the solution
5 Account for the following:
   a) pH of the medium changes enormously at its equivalence point.
   b) At half equivalence point, pH is equal to pKa.
6 Write down Henderson-Hasselbalch equation.
7 pKa values of Acetic acid and Formic acid are 4.7 and 3.7 respectively at a given temperature. Which one of the two acids is stronger?
8 Henderson-Hasselbalch equation is holds good for buffer of weak acid and its salt with strong base. How do would you justify the use of this equation in the experiment?
9 What is a buffer solution? Give any two examples.
10 What are the applications of Henderson-Hasselbalch equation?
11 Find out the ratio in which the acid and its salt are to be dissolved in 1 dm^3 such that the resultant buffer would offer a P^H of 5.95. Given pKa = 4.0.
12 1.0 mole of sodium acetate and 0.5 mole of Acetic acid were dissolved in 1 dm^3 of water. Calculate the pH of the buffer. Given, dissociation constant of Acetic acid at 298 K is 10^{-5}
13 Why is half equivalence point taken into account for assigning pKa?

6. Colorimetric estimation of copper

Draw 2,4,6,8 and 10 cm^3 of given copper sulphate solution into separate 25 cm^3 volumetric flasks. Add 2.5 cm^3 of ammonia to each of them and also into the test solution of unknown concentration. Dilute with water up to the mark and mix well. Measure the absorbance (λ_{max}**= 620 nm**) of each of these against blank solution (Only ammonia and water). Plot a graph of absorbance (OD) against volume of $CuSO_4$ and determine the concentration of copper in the test solution.



VIVA QUESTIONS

1 What is Colorimetry?
2 What forms the basis for Colorimetric determination?
3 What is photoelectric Colorimeter?
4 What are filters? Why are they used?
5 What is wavelength?
6 What is frequency?
7 Why is estimation of copper done at 620 nm wave length?
8 What is wave number?
9 State Beer's law.
10 State Lambert's law.
11 State Beer-Lambert's law.
12 Upon what factors the absorbance of a solution depends?
13 What is calibration curve?
14 What is meant by transmittance?
15 Why is a filter of 620 nm used in the estimation of copper?
16 Mention a few important criteria for satisfactory Colorimetric analysis.
17 Mention a few advantages of Photoelectric Colorimetric determination.
18 What is Blank solution?
19 What is the basic requirement of a compound for its estimation by Colorimetric method?
20 Why is Ammonia added? Why is that same amount of Ammonia added?

7. Determination of viscosity coefficient of a given liquid using Ostwald's viscometer

Pipette out 10 cm^3 of the given liquid into the wide limb of the dry viscometer kept in a water bath. Suck the liquid through the other limb. Determine the time of flow between the two fixed points, one above and one below the bulb in the narrow limb of viscometer. Repeat and calculate the average time of flow. Pour out the liquid and wash the viscometer with acetone and dry it. Pipette out 10 cm^3 of distilled water into the wide limb, repeat the experiment and determine the average time of flow. From the densities of liquid and water, viscosity coefficient of water, determine the viscosity coefficient of the liquid.

$$\eta_l = \frac{t_l\, d_l\, \eta_w}{t_w\, d_w}$$

(1 milli poise = $10^{-4} Nsm^{-2}$)

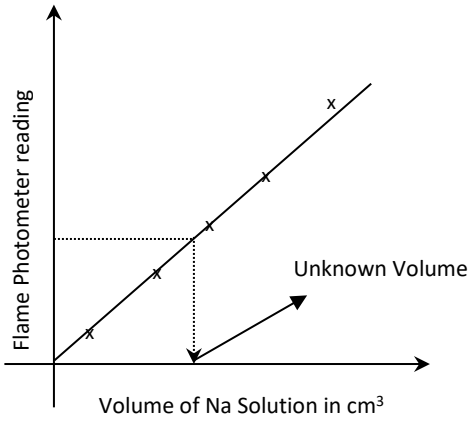
Where,
d_l = density of liquid;
d_w = density of water
t_l time of flow of liquid;
t_w = time of flow of water
η_w = viscosity of water

VIVA QUESTIONS

1 What is Viscosity Coefficient of a liquid?
2 What is density of a liquid?
3 What is specific gravity?
4 How are specific gravity and density related?
5 What is SI unit of Viscosity Coefficient?
6 What are the factors that affect the viscosity of a liquid?
7 How does the viscosity vary with temperature?
8 Why is acetone used for cleaning viscometer?
9 Why do you require laboratory temperature for viscosity determination?
10 How is the viscosity of liquid related to its mobility?
11 What is fluidity of a liquid?

8. Estimation of Sodium in the given solution using flame photometer

Draw 2,4,6,8 and 10 cm^3 of given Sodium chloride solution into separate 25 cm^3 volumetric flasks. Dilute with water up to the mark and mix well. Similarly, prepare the test solution of unknown concentration. Note down the flame photometer readings (**λ_{max} = 586 nm**). Plot a graph of flame photometer readings against the volume of the solution get the calibration curve. Using the curve obtained find out the volume of the unknown solution containing sodium and calculate the amount of sodium in it.

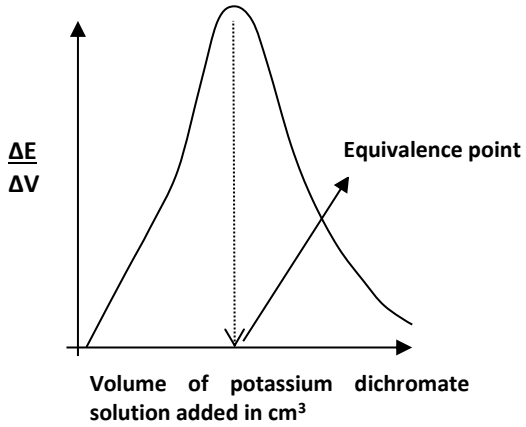


VIVA QUESTIONS

1. What is meant by atomic emission?
2. What is Flame Photometry?
3. What are alkali and alkaline earth metals?
4. What are the various events that occur when solution containing metal atom ions atomized through a flame?
5. What are different components of a Flame Photometer?
6. Why the analysis of Sodium is advantageous is Flame Photometry?
7. What is the role of filter in Flame Photometry?
8. What are the errors that occur in Flame Photometry?
9. What are the factors that influence this experiment?
10. Name the fuel and oxidant used in this experiment

9. Estimation of Ferrous ion by Potentiometric titration method

Pipette out 25cm^3 of the given ferrous ammonium sulphate solution into a beaker. Add 1 test tube of dil.H_2SO_4. Immerse the platinum electrode and calomel electrode assembly into it. Connect the electrodes to potentiometer and measure the potential. Add potassium dichromate solution from the burette in the increments of 0.2 cm^3 measure the potential after each addition. Plot a graph of ΔE/ΔV against volume of potassium dichromate solution added and determine the equivalence point. From normality of potassium dichromate evaluate the normality of given ferrous ammonium sulphate.

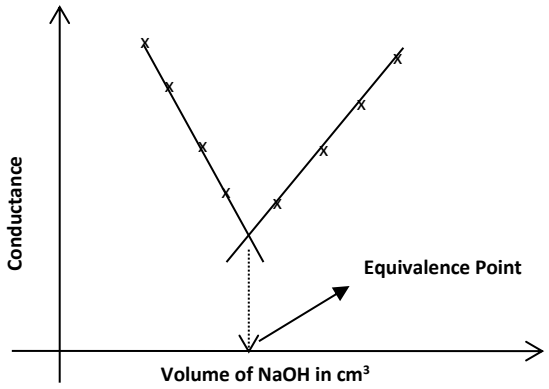


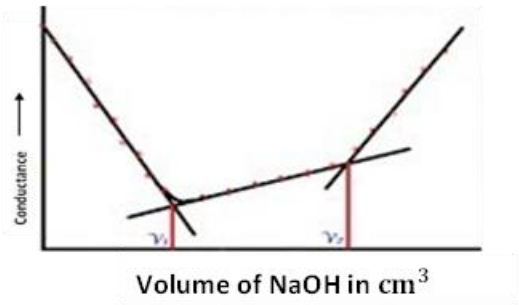
VIVA QUESTIONS

1. What is a Potentiometer titration?
2. Give the principle of Potentiometer titration.
3. What are the electrodes used in Potentiometer titration?
4. What is determining factor in the oxidation-reduction reaction?
5. What is an indicator electrode?
6. What is the reaction that occurring between FAS and Potassium Dichromate?
7. What are the advantages of Potentiometric titration?

10. Conductometric estimation of Hydrochloric acid

Pipette out 25cm^3 of HCl in to a beaker. Immerse the conductivity cell in it. Connect the cell to the conductivity bridge and measure the conductance. Add sodium hydroxide in the increments of 0.2cm^3 and measure the conductance after each addition. Plot a graph of conductance versus volume of sodium hydroxide added. Determine the volume of sodium hydroxide required to neutralize HCl. From the normality of sodium hydroxide, calculate the normality and amount of HCl.





VIVA QUESTIONS

1 State Ohm's law.
2 What is conductance?
3 What is the unit of conductance?
4 Name the SI unit of conductance.
5 Mention the different types of conductivities.
6 Which of the above conductivity measured during Conductometric titration?
7 What is specific conductivity?
8 What is equivalent conductivity?
9 What is molar conductivity?
10 What is a cell?
11 Define cell constant
12 Describe a conductivity cell.
13 What is the principle involved in Conductometric titration?
14 How is the accuracy of the method determined?
15 Explain the graph obtained in case of Conductometric titration involving acid mixture and NaOH.
16 What are the advantages of Conductometric titration over visual or potentiometric titration?
17 How do you account for the decrease in the conductivity at the beginning of the titration?
18 How would you account for the increase in the conductivity after the end point?
19 The second line in the graph is not as steep as the first. Explain.

11. To find Tg of Polymer using DSC

A sample of 1-4 mg of polymer sample is placed in a 40 µl aluminum pan. Adapt the sample size to your pan size, if different. The polymer sample should cover the bottom of the pan. The filled pan is crimped and locked with Aluminium lid. DSC Instrument is kept on for 30 minutes to attain the required temperature. The pan containing sample is placed in the heating chamber with extreme care. Give the required inputs like heating rate, sample weight, cooling time etc in the software. Run the DSC machine and get the heat versus temperature graph. The proper interpretation of the obtained graph will give you the glass transition temperature of the sample measured.

The glass transition results in a kink in the heat versus temperature plot due to the change in heat capacity (A). In a plot of heat flow versus temperature it is a gradual transition that occurs over a range of temperatures (B). The glass transition temperature is taken to be the middle of the sloped region.

VIVA QUESTIONS

1. What is Glass transition temperature?
2. What is the significance of Tg?
3. What are factors effecting Tg?
4. What is the principle of this experiment?
5. Why Glass transition temperature is in range?
6. Explain the interpretation of Graph A and B

12. Determination of relative and Kinematic Viscosities of given lubricating oil at different temperatures using Redwood Viscometer

Clean the viscometer cup properly with the help of suitable solvent e.g. CCl4, ether, petroleum spirit or benzene and dry it to remove any traces of solvent. Level the viscometer with the help of leveling screws and fill the outer bath with water for determining the viscosity at 80 oC and below. Place the ball valve on the jet to close it and pour the test oil into the cup up to the tip of indicator. Place a clean dry Kohlrausch flask immediately below and directly in line with discharging jet. Insert a clean thermometer and a stirrer in the cup and cover it with a lid. Heat the water filled in the bath slowly with constant stirring. When the oil in the cup attains a desired temperature, stop the heating. Lift the ball valve and start the stopwatch, oil from the jet flows into the flask. Stop the stopwatch when lower meniscus of the oil reaches the 50 ml mark on the neck of receiving flask. Record the time taken for 50 ml of the oil to collect in the flask. Repeat the experiment to get more readings.

VIVA QUESTIONS

1. What is Viscosity?
2. What are different types of viscosity explain them and write the units?
3. What are factors effecting viscosity?
4. Mention some applications where viscosity is considered?
5. Relation between density and viscosity?
6. Which constructional feature of viscometer varies with the viscosity of oil?

13. Preparation of ZnO by Solution Combustion method

Weigh exactly 5.0 g of Magnesium Nitrate and 1.95 g of Urea (1:1 stoichiometry) in 250 ml beaker and dissolve the salts by adding minimum quantity of distilled water and stirring on a magnetic stirrer for 10 min. The uniformly dissolved redox mixture is kept in a pre-heated furnace maintained at 500 ºC. The redox solution undergoes evaporation and finally results in viscous gel containing uniformly mixed Magnesium Nitrate and Urea. After some time, it catches fire with the liberation of gases and heat energy, finally results in highly porous, white powder of nano MgO.

VIVA QUESTIONS

1.What is the formula for the calculation of amount of fuel?
2.What is the principle behind Solution Combustion method?
3.What are the advantages of Solution Combustion method?

14. Electroplating of copper on metallic objects

Pour 200 ml of the electrolyte solution into the beaker. Electrolyte solution is a strong solution of copper sulfate plus a few drops of sulfuric acid to keep the solution acidic. Attach connecting wires with alligator clips to the terminals of the power supply or a 6-volt battery. Clean the coins with a mixture of 3 g NaCl and 15 ml vinegar: rinse and dry. Tightly wrap one end of a 10 cm length of copper around each coin, leaving 5-6 cm of wire free. Mass each coin-copper wire assembly and record the masses. Push the free end of each wire through the cardboard square and place the square over the beaker so that the coin “electrodes” are immersed in the electrolyte solution as illustrated below. **Note**: the two electrode assemblies must not touch. Attach the connecting wires to the top of the copper wire assemblies. Allow the electroplating cell to operate for 30-60 minutes. Record the exact time the cell was operating (optional). Record the ammeter reading from the meter or power supply (optional). Remove each “electrode;” dry, being careful not to lose any of the copper plating; mass each and record.

VIVA QUESTIONS

1. What is electroplating?
2. Write down the cathodic reactions involved in electroplating of Cu on a coin.
3. List out 3 factors on which electroplating depends.
4. What is throwing power?
5. Highlight the difference between electroplating and electroless plating.
6. What is chrome plating?
7. Why isn’t Chromium anode used in chrome plating?
8. Mention at least two applications of electroplating of copper and chromium.